

ANTIA
LAMAS-LINARES
QUANTUM SCIENTIST

M	Available on request
@	Available on request
W	Google Scholar
CV	PDF version

SUMMARY

Experimental quantum information scientist with a track record of defining and driving product roadmaps across business units and teams. Most recently led the integration of quantum technologies into AWS networking infrastructure, building trusted connections between research teams, engineering organizations, business development, legal, and finance. Technical work centers on quantum communication protocols, QKD systems, real-world security vulnerabilities, satellite-based quantum communications, and quantum-enabled optical ground stations.

WORK EXPERIENCE

PRINCIPAL APPLIED QUANTUM SCIENTIST

Amazon Web Services | 2021-2026 | USA

- Initiated quantum networking activities at AWS and established the AWS Center for Quantum Networking.
- Transitioned QKD from a research-oriented quantum organization into AWS network infrastructure.
- Filed seven patents in QKD, secure randomness generation, hybrid terrestrial and space quantum communications, quantum clock synchronization, and entanglement generation.
- Oversaw collaborations and proofs of concept with HSBC, Fidelity, and Singapore NQSN.

VISITING PROFESSOR

Vigo Quantum Communications Center | 2023-present | Spain

- Lead the Satellite Quantum Communications Group.
- Supervise the design and build-up of a quantum-enabled optical ground station.

ADVISORY BOARD MEMBER

Integrated Quantum Networks Hub | 2025-present | UK

- Advise the UK IQN Hub, a main funding mechanism for UK research in quantum networking.

CHIEF QUANTUM SCIENTIST

SpeQtral | 2019-2020 | Singapore

- Provided scientific strategic input for a startup focused on space-based quantum communications technology.

HPC RESEARCHER

Texas Advanced Computing Center | 2015-2019 | Austin

- Led TACC quantum computing efforts and worked on scheduler optimization for heterogeneous systems.

ACADEMIC AND RESEARCH LEADERSHIP

NUS, CQT, NIST, UCSB | 2002-2020 | Singapore, USA

- Held roles including Assistant Professor at NUS, founding PI at CQT, Senior NIST-ARRA Fellow, Visiting Associate Research Professor, and Postdoctoral Researcher at UC Santa Barbara.
- Member of the senior CQT team that secured over \$100M in funding.

SELECTED PUBLICATIONS AND PATENTS

Full list at [google scholar profile](#)

PATENTS

- Quantum Safe Enclaves, US Patent 12,615,140, granted 2026.
- Metered Distribution of Quantum Entanglement, US Patent 12,489,612, granted 2025.
- Hybrid space-fiber quantum networks, US Patent 11,962,354, granted 2024.
- Quantum secure network clock synchronization, US Patent 12,143,203, granted 2024.

PAPERS

- Full-field implementation of a perfect eavesdropper on a quantum cryptography system, Nature Communications, 2011.
- Experimentally faking the violation of Bell's inequalities, Physical Review Letters, 2011.
- Experimental quantum cloning of single photons, Science, 2002.
- Stimulated emission of polarization-entangled photons, Nature, 2001.



EDUCATION

PHD IN PHYSICS

University of Oxford, UK

MSC IN APPLIED OPTICS

Imperial College, UK

LICENCIATURA IN PHYSICS

Universidade de Santiago, Spain

SKILLS

TECHNICAL

- Quantum communications systems
- QKD and quantum hacking
- Quantum time transfer
- Cryogenic single-photon detectors
- Entangled photon sources
- Satellite quantum links

LEADERSHIP

- Technical roadmaps
- C-level strategy engagement
- Experimental facility setup
- International collaborations

LANGUAGES

- English: fluent
- Spanish: native
- Portuguese: advanced
- German: basic